

Week 1 Homework (March 3)

Due date: Mar 10 in class

1. Let $P_{\mathbf{v}_1, \dots, \mathbf{v}_k}$ be the parallelepiped spanned by the vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in the real vector space V equipped with inner product g , prove that

$$\text{Vol}_g(P_{\mathbf{v}_1, \dots, \mathbf{v}_k}) = \sqrt{\det(\langle \mathbf{v}_i, \mathbf{v}_j \rangle_g)}.$$

2. Use the *Spectral Theorem* to prove the *Classification of symmetric Bilinear forms* on real vector space of dimension n .
3. Let V be a real vector space, and W be a subspace, define a binary relation:

$$\mathbf{v} \sim_W \mathbf{v}' \text{ iff } \mathbf{v} - \mathbf{v}' \in W.$$

- (a) Prove that “ $\sim_W$ ” is an equivalence relation on V .
- (b) Define a scalar multiplication and addition on $V/\sim_W$ via

$$a\alpha + b\beta := [a\mathbf{u} + b\mathbf{v}]_{\sim_W}$$

where $\mathbf{u} \in \alpha, \mathbf{v} \in \beta$ for any equivalence classes α and β . Show that $[a\mathbf{u} + b\mathbf{v}]_{\sim_W}$ is independent of the particular choice of $\mathbf{u}$ and $\mathbf{v}$ (this is called the “well-definedness” of the operation). This makes $V/\sim_W$ a vector space, called the quotient space, which is also denoted by V/W .